Langevin Autoencoders for Learning Deep Latent Variable Models

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Motivation

Learning deep latent variable models

Latent variable models

$$p(\mathbf{x};\boldsymbol{\theta}) = \int p(\mathbf{x} \mid z;\boldsymbol{\theta})p(z)dz$$

Gradient over model parameters

$$\nabla_{\boldsymbol{\theta}}\mathbb{E}_{\hat{p}_{\text{data}}(\mathbf{x})}[\log p(\boldsymbol{x};\boldsymbol{\theta})] \approx \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}_{p(\mathbf{z}^{(i)} \mid \boldsymbol{x}^{(i)};\boldsymbol{\theta})}\left[\nabla_{\boldsymbol{\theta}}\log p\left(\boldsymbol{x}^{(i)},\boldsymbol{z}^{(i)};\boldsymbol{\theta}\right)\right]$$

- To learn a latent variable model via gradient ascent, we need to approximate the expectation over the latent posterior $p(\mathbf{z} \mid \mathbf{x};\boldsymbol{\theta})$, which is intractable in general.
- Variational inference is a dominant approach of approximation (e.g., VAE [1]), but the approximation power is limited.
- Markov chain Monte Carlo (MCMC) is capable of approximate complex posteriors, but is too slow due to datapoint-wise iterations.

Preliminary

Langevin dynamics

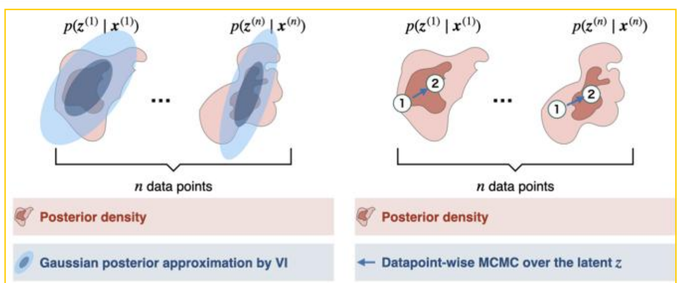
- Langevin dynamics [2] is a gradient-based MCMC.

$$z' \sim q\left(z' \mid z\right) = \mathcal{N}\left(z'; z + \eta\nabla_z \log p(\boldsymbol{x}, z;\boldsymbol{\theta}), 2\eta\boldsymbol{I}\right)$$

$$z \leftarrow z' \text{ with probability } \min\left\{1, \frac{p\left(\boldsymbol{x}, z';\boldsymbol{\theta}\right)q\left(z \mid z'\right)}{p\left(\boldsymbol{x}, z;\boldsymbol{\theta}\right)q\left(z' \mid z\right)}\right\}$$

- By repeating this iterative update, the samples asymptotically approach to the true posterior $p(\mathbf{z} \mid \mathbf{x};\boldsymbol{\theta})$.
- MCMC iterations are independently performed for each posterior per datapoint $p(\mathbf{z}^{(i)} \mid \boldsymbol{x}^{(i)};\boldsymbol{\theta})$ for $i = 1,\ldots,n$.

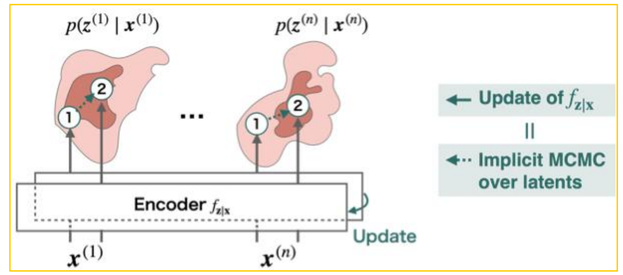
➡How can we amortize the cost of datapoint-wise iterations?



(a) Variational inference (b) Langevin dynamics

Method

Amortized Langevin dynamics



- In our amortized Langevin dynamics (ALD), we prepare an encoder that maps the observed data into the latent variable, and run MCMC on its parameter space.
- Markov chain on the latent space is implicitly performed by collecting the output of the encoder.
- Our main theorem shows that the ALD is valid as MCMC if the encoder takes the form of:

$$f_{\mathbf{z}|\mathbf{x}}(\boldsymbol{x};\boldsymbol{\Phi}) = \Phi g(\boldsymbol{x}).$$

- It can be easily implemented using a neural net whose parameters are fixed except for the last linear layer.

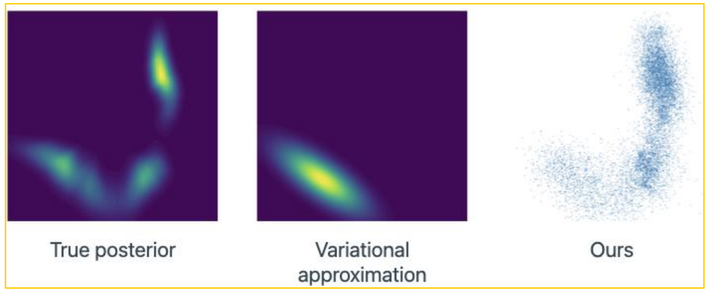
Langevin Autoencoder

- ALD can be applied to the training of deep latent variable models (DLVMs) by slightly modifying the learning algorithm of traditional autoencoders.
- We call this learning algorithm of DLVMs the ***Langevin autoencoder (LAE)***.

$\boldsymbol{\theta}, \boldsymbol{\Phi}, \boldsymbol{\psi} \leftarrow$ Initialize parameters
repeat
  $\boldsymbol{x}^{(1)},\ldots,\boldsymbol{x}^{(n)} \sim \hat{p}\left(\mathbf{x}\right)$
  for $t = 0,\ldots,T-1$ **do**
    $V_t = -\sum_{i=1}^{n}\log p\left(\boldsymbol{x}^{(i)}, \boldsymbol{z}^{(i)} = \boldsymbol{\Phi}g\left(\boldsymbol{x}^{(i)};\boldsymbol{\psi}\right);\boldsymbol{\theta}\right)$
    $\boldsymbol{\Phi}' \sim q\left(\boldsymbol{\Phi}' \mid \boldsymbol{\Phi}\right) := \mathcal{N}\left(\boldsymbol{\Phi}';\boldsymbol{\Phi} - \eta\nabla_{\boldsymbol{\Phi}}V_t, 2\eta\boldsymbol{I}\right)$
    $V_t' = -\sum_{i=1}^{n}\log p\left(\boldsymbol{x}^{(i)}, \boldsymbol{z}^{(i)} = \boldsymbol{\Phi}'g\left(\boldsymbol{x}^{(i)};\boldsymbol{\psi}\right);\boldsymbol{\theta}\right)$
    $\boldsymbol{\Phi} \leftarrow \boldsymbol{\Phi}'$ with probability $\min\left\{1, \frac{\exp\left(-V\left(\boldsymbol{\phi}'\right)\right)q\left(\boldsymbol{\phi} \mid \boldsymbol{\phi}'\right)}{\exp\left(-V\left(\boldsymbol{\phi}\right)\right)q\left(\boldsymbol{\phi}' \mid \boldsymbol{\phi}\right)}\right\}$
  end for
  $V_T = -\sum_{i=1}^{n}\log p\left(\boldsymbol{x}^{(i)}, \boldsymbol{z}^{(i)} = \boldsymbol{\Phi}g\left(\boldsymbol{x}^{(i)};\boldsymbol{\psi}\right);\boldsymbol{\theta}\right)$
  $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta\nabla_{\boldsymbol{\theta}}\frac{1}{T}\sum_{t=1}^{T}V_t$
  $\boldsymbol{\psi} \leftarrow \boldsymbol{\psi} - \eta\nabla_{\boldsymbol{\psi}}\frac{1}{T}\sum_{t=1}^{T}V_t$
until convergence of parameters
return $\boldsymbol{\theta}, \boldsymbol{\Phi}, \boldsymbol{\psi}$

Results

Toy example



- We perform posterior inference for a randomly initialized DLVM using variational inference (VI) and our ALD.
- VI fails to capture the multimodality of the true posterior.
- Our ALD can approximate complex multimodal posteriors in toy examples.

Image density estimation

	MNIST	SVHN	CIFAR-10	CelebA
VAE	1.189 ± 0.002	4.442 ± 0.003	4.820 ± 0.005	4.671 ± 0.001
VAE-flow	1.183 ± 0.001	4.454 ± 0.016	4.828 ± 0.005	4.667 ± 0.005
Hoffman (2017) [3]	1.189 ± 0.002	4.440 ± 0.007	4.831 ± 0.005	4.662 ± 0.011
LAE (ours)	$\mathbf{1.177} \pm 0.001$	$\mathbf{4.412} \pm 0.002$	$\mathbf{4.773} \pm 0.003$	$\mathbf{4.636} \pm 0.003$

Likelihood evaluation for test data

- Our LAE consistently outperforms VI-based methods such as the VAE and other existing MCMC-based methods.

Future works

- How can we remove the bias of finite-step MCMC samples?
  - Recently proposed unbiased MCMC methods [4] may be useful.
- Is it possible to apply the ALD (and LAE) to the state-of-the-art DLVMs, such as NVAE [5]?

References

[1] Diederik P Kingma and Max Welling (2013). "Auto-encoding variational bayes." In: arXiv preprint arXiv:1312.6114.

[2] Radford M Neal (2011). "Mcmc using hamiltonian dynamics." In: Handbook of Markov Chain Monte Carlo, page 113.

[3] Matthew D Hoffman (2017). "Learning deep latent gaussian models with markov chain monte carlo." In: International conference on machine learning, pages 1510–1519. PMLR.

[4] Pierre E Jacob, John O'Leary, and Yves F Atchadé (2020). "Unbiased markov chain monte carlo methods with couplings." In: Journal of the Royal Statistical Society: Series B (Statistical Methodology), 82(3): 543–600.

[5] Arash Vahdat and Jan Kautz (2020). "Nvae: A deep hierarchical variational autoencoder." In: Advances in Neural Information Processing Systems, 33:19667–19679.